

QUANTITATIVE ANALYTICS · MBA DISSERTATION

Bank Loan Default Prediction

Research Questions, Methods, and Answers

The Problem

Lenders need to decide, quickly and at scale, whether a borrower will repay. Traditional credit scores and manual underwriting are backward-looking and miss interaction effects between risk factors.

This study asked three questions:

- 1 Which factors actually drive loan default?
- 2 Can we reliably predict individual default risk?
- 3 Does the timing of default vary by borrower type?

44,541

loan records analysed

22.32%

overall default rate

14

borrower & loan variables

3

ML models compared

QUESTION 1

Which factors actually drive loan default?

PROCESS

1

Formed 12 hypotheses

e.g. does interest rate, income, or home ownership differ between defaulters and repayers?

2

Chose the right test per variable

Welch t-test (continuous), Chi-square (categorical), Mann-Whitney U (non-normal)

3

Tested at 5% significance

Compared defaulters vs. non-defaulters across all 14 variables

ANSWER

9 of 12 hypotheses rejected

Default is driven by financial strain, not demographics:

- Interest rate — $t = 74.54$, $p < .001$
- Loan-to-income ratio — $t = 88.28$, $p < .001$
- Prior default history — $\chi^2 = 13,173.5$, $p < .001$
- Home ownership (renters riskiest) — $p < .001$

Not significant: credit score, education, gender

QUESTION 2

Can we reliably predict individual default risk?

PROCESS

1

Split data 80/20

35,632 train / 8,909 test, stratified to preserve the 22.3% default rate

2

Trained 3 classifiers

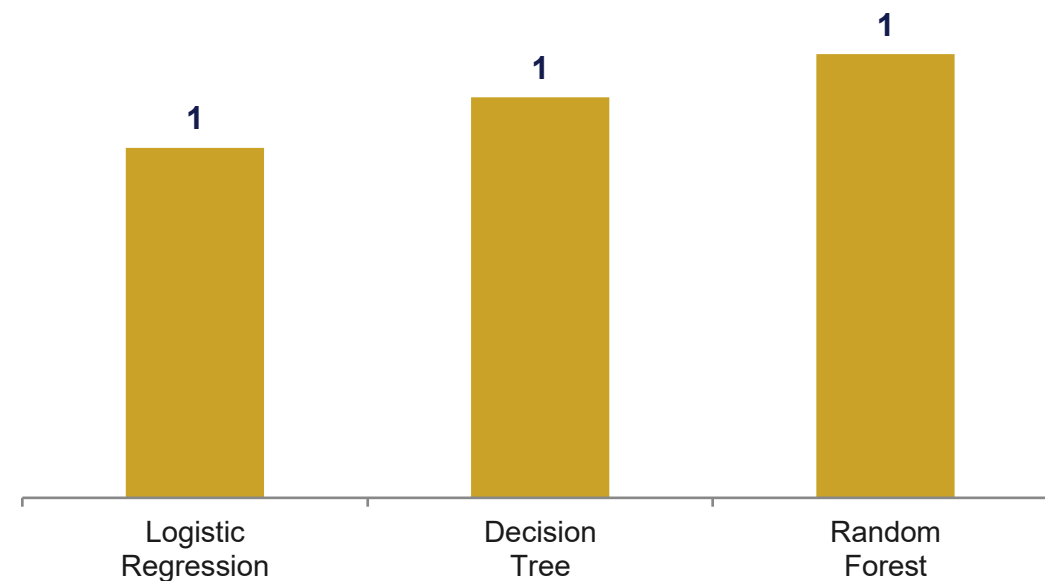
Logistic Regression, Decision Tree, Random Forest — all class-weight balanced

3

Compared on ROC-AUC

Also checked recall, F1-score, and segmented applicants into risk tiers

ANSWER — Model Comparison (ROC-AUC)



Random Forest wins: ROC-AUC 0.973, 92.9% accuracy. Applicants segmented into High (29.7%) / Medium (11.0%) / Low (59.3%) risk.

QUESTION 3

Does the timing of default vary by borrower type?

PROCESS

1

Applied Cox Proportional Hazards

Modelled time-to-default using credit history length as the duration variable

2

Treated non-defaulters as censored

34,599 of 44,541 borrowers had not defaulted — handled as right-censored data

3

Compared hazard ratios

Interest rate, rental housing, prior default, and employment experience as covariates

ANSWER

Yes — high-risk borrowers default earlier, not just more often

Interest rate

37.5% higher hazard per SD increase

Renters

31.2% higher hazard than non-renters

Employment experience

67% lower hazard — strongest protective factor

Concordance Index (C-index): 0.927 — excellent temporal discrimination

The Key Finding

Loan default is a financial affordability problem — not a demographic one.

Credit score

631.85 vs 632.70

Virtually identical between defaulters and non-defaulters

Debt-to-income ratio

2.4x more predictive

Engineered feature outranks raw credit score in the model

Education & gender

Ranked 13th & 14th

No meaningful predictive signal among all 14 features

Recommendation: replace credit-score-first underwriting with debt-to-income and prior-default screening.

Thank You

Full dissertation, Python notebooks, and Power BI dashboard available in the below link (in pdf format) .

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